

AWS Gateway Load Balancer — Teardown Procedure

Step-by-step removal of GWLB endpoints, endpoint service, and load balancer infrastructure

Platform: AWS **Service:** VPC, EC2 (ELBv2) **Risk Level:** High — production traffic impact

WARNING — TRAFFIC IMPACT

This procedure removes inline security inspection infrastructure. All traffic flowing through the GWLB will be affected. Ensure you have a maintenance window and a rollback plan before starting. Perform in **non-production environments first**.

Dependency Chain

GWLB components have strict dependency ordering. Deleting a component while dependents still reference it will either fail (API error) or create blackhole routes that silently drop traffic. Always delete from the outermost consumer inward.

DELETION ORDER (LEFT TO RIGHT)

Routes → GWLB Endpoints → Endpoint Service → Listeners → Target Group →
GWLB

PHASE 1

Remove Routes (Consumer VPCs)

Before touching any GWLB components, update every route table that sends traffic to a GWLB endpoint. Check **all consumer/spoke VPCs** — routes may span multiple VPCs and availability zones.

Steps

1. **Identify all route tables** referencing GWLB endpoints:

VPC Console → Route Tables

Filter by target type `vpce-*`, or search for the specific endpoint IDs. Document every match.

2. **Delete or replace each route:**

- If traffic should stop entirely: **delete the route**
- If traffic should bypass inspection temporarily: **replace the target** (e.g., point to IGW or NAT Gateway)

3. **Verify:** Confirm no route table still references any of the GWLB endpoint IDs

WHY ROUTES FIRST?

If you delete endpoints before removing routes, the routes become **blackhole routes**. Traffic silently drops instead of failing visibly. Removing routes first makes the traffic path explicit and prevents silent outages.

PHASE 2

Delete GWLB Endpoints (Consumer VPCs)

These are VPC endpoints of type `GatewayLoadBalancer` in the consumer/spoke VPCs. There is typically one per AZ per consumer VPC.

Steps

1. **List all GWLB endpoints:**

VPC Console → Endpoints

Filter: Type = `GatewayLoadBalancer` . Note which endpoint service each connects to.

2. **Delete each endpoint:**

Select endpoint → **Actions** → **Delete VPC Endpoints** → Confirm

3. **Wait** for endpoint state to show `deleted` (typically 1–2 minutes)

WHY BEFORE ENDPOINT SERVICE?

The endpoint service will refuse deletion if any endpoints are still connected to it. All consumer-side endpoints must be removed first.

PHASE 3

Delete the Endpoint Service (Security VPC)

This is the VPC endpoint service in the security/appliance VPC that exposes the GWLB to consumer VPCs.

Steps

1. Navigate to VPC Console → Endpoint Services

2. Select the endpoint service tied to your GWLB

3. Check the **Endpoint Connections** tab — it should show no active connections (empty after Phase 2)

4. **Actions** → **Delete Endpoint Service** → Confirm

STILL SHOWING CONNECTIONS?

If the Endpoint Connections tab shows remaining endpoints, return to Phase 2 and delete them. The service will not allow deletion while connections exist.

PHASE 4**Delete GWLB Listeners****Steps**

1. Navigate to `EC2 Console → Load Balancers`
2. Select the Gateway Load Balancer
3. Go to the **Listeners** tab
4. Select each listener → **Actions** → **Delete listener**

NOTE

GWLB typically has a single listener (IP protocol on port 6081 — GENEVE). Verify all listeners are removed before proceeding.

PHASE 5**Delete the Target Group****Steps**

1. Navigate to `EC2 Console → Target Groups`
2. Select the target group associated with the GWLB
3. **Deregister all targets:** Targets tab → select all → **Deregister**
4. Wait for target states to transition: `draining` → `unused`
5. **Actions** → **Delete target group**

NOTE

You can delete the target group without explicitly deregistering targets first, but explicit deregistration is cleaner and avoids edge cases with lingering health-check traffic.

PHASE 6**Delete the Gateway Load Balancer****Steps**

1. Navigate to `EC2 Console → Load Balancers`
2. Select the Gateway Load Balancer
3. **Actions** → **Delete load balancer**
4. Type `confirm` in the confirmation dialog → Delete
5. Wait for the GWLB to disappear from the load balancer list

Post-Removal Verification

After completing all six phases, verify no orphaned resources remain:

- ☐ No route tables reference `vpce-*` targets that no longer exist
- ☐ No orphaned ENIs remain in the GWLB subnets
`EC2 Console → Network Interfaces` — filter by subnet
- ☐ Security groups used exclusively by the GWLB are deleted (or documented for later cleanup)
- ☐ If GWLB subnets are no longer needed, they can be removed
- ☐ CloudWatch alarms referencing the GWLB or target group are removed
- ☐ Any cross-account endpoint service permissions (allowed principals) have been cleaned up

AWS CLI Reference

If performing this via CLI instead of the console, the commands follow the same order:

PHASE	CLI COMMAND
1 — Routes	<code>aws ec2 delete-route --route-table-id rtb-xxx --destination-cidr-block 0.0.0.0/0</code>

PHASE	CLI COMMAND
2 — Endpoints	<code>aws ec2 delete-vpc-endpoints --vpc-endpoint-ids vpce-xxx</code>
3 — Endpoint Service	<code>aws ec2 delete-vpc-endpoint-service-configurations --service-ids vpce-svc-xxx</code>
4 — Listeners	<code>aws elbv2 delete-listener --listener-arn arn:aws:...</code>
5 — Target Group	<code>aws elbv2 delete-target-group --target-group-arn arn:aws:...</code>
6 — GWLB	<code>aws elbv2 delete-load-balancer --load-balancer-arn arn:aws:...</code>

TIP

Use `aws ec2 describe-route-tables --filters "Name=route.gateway-id,Values=vpce-*"` to find all route tables referencing GWLB endpoints across a region. Repeat for each region where GWLB infrastructure was deployed.